

Alfaj Uddin Ahmed

PH.D. STUDENT (BIOMEDICAL ENGINEERING) & M.S. STUDENT (ELECTRICAL & COMPUTER ENGINEERING), PURDUE UNIVERSITY

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West Lafayette, IN, United States

RESEARCH INTERESTS

Large Language Models | Reasoning and Long-Horizon Execution Failures | Vision-Language Models, Grounding, and Hallucination | Evaluation and Failure-Mode Analysis | Generative AI and Diffusion Models | Parameter-Efficient Fine-Tuning | Clinical NLP | Medical Image Analysis | Rare-Event Detection under Label Scarcity | Scalable ML Systems

EDUCATION

- **Purdue University — Leslie A. Geddes Fellow** Aug 2025 – Present
Ph.D. in Biomedical Engineering — Computational Medical Imaging and Machine Learning
◦ **Dissertation Area:** ML for pediatric fracture and abuse detection from radiographs; LLM and VLM reliability. West Lafayette, IN
- **Purdue University** Aug 2026 – Present
M.S. in Electrical and Computer Engineering — concurrent with the Ph.D. West Lafayette, IN
- **Bangladesh University of Engineering and Technology (BUET)** May 2017 – May 2022
B.Sc. in Biomedical Engineering — GPA: 3.43/4.00
◦ **Undergraduate Thesis:** Automated COVID-19 infection detection and segmentation from lung CT volumes using UNet++ and 3D convolutional networks [slides]. Dhaka, Bangladesh

RESEARCH EXPERIENCE

- **Purdue University** Aug 2025 – Present
Leslie A. Geddes Fellow and Graduate Research Assistant — Weldon School of Biomedical Engineering West Lafayette, IN
 - **Long-Horizon LLM Execution (submitted to ICLR 2027).** Showed that benchmark accuracy hides *how* LLMs fail on multi-step deterministic tasks: built an exact-simulator evaluation that separates correct responses from decision errors, state errors, token exhaustion, malformed outputs, and noncompliance, run as 800-sample reasoning-on/off grids on **Qwen3** models served through **vLLM** on dual-GPU nodes, with Clopper–Pearson and clustered uncertainty estimates.
 - Found that failure composition shifts with scale without a monotonic law: smaller models choose wrong moves while larger models choose correct moves but misreport the resulting state; reasoning mode collapsed move-selection errors from roughly 60% to 0% while leaving state-transcription errors unchanged, and position labels cut Qwen3-4B state errors from 50.6% to 0.6% while adding formatting failures. Replicated on Tower of Hanoi.
 - Built the harness behind those results: 810 automated tests, SHA-256-pinned verbatim prompts, parser audits, token-budget controls, and resumable detached grid runners for unattended multi-day GPU sweeps.
 - **Vision-Language Grounding (manuscript in final preparation for CVPR 2027).** Characterized the “mirage” failure in which VLMs answer confidently from blank or unrelated images, and evaluated **TIDE**, a training objective pairing each question with its correct image, a blank image requiring refusal, and a same-domain wrong image requiring refusal, with a population-level guarantee that image-blind predictors are suboptimal by at least log 3 nats.
 - Fine-tuned **Qwen3-VL** at 2B/4B/8B/32B with **4-bit QLoRA**; TIDE was evaluated with LoRA on six VLMs scored by an image-aware judge: blank-image mirage fell to zero on all six and between-model variation on unseen out-of-domain images narrowed from 0.473 to 0.042, while in-domain wrong-image mirage persisted at 27.8–34.8%.
 - Traced an apparent scale-dependent trend across the model ladder to a projector co-scaling artifact, reporting a documented null result; showed that representation-level grounding alone does not reduce blank-image mirage on converged models.
 - **Dissertation — Generative Augmentation for Rare-Lesion Detection.** Designing a **mask-conditional latent diffusion model** that operates in a VQ-VAE latent space with **ControlNet**-style conditioning on radiologist-delineated lesion masks, for a regime with only a handful of confirmed abuse cases per center per year; the pipeline couples diffusion augmentation with transformer detection and few-shot classification, after a pilot showed unconditioned diffusion cannot produce coherent skeletal anatomy.
 - **Neuromorphic Sensing Interfaces (under review at ICASSP 2027).** Showed that rate readout from leak-free spiking front-ends on event-camera data discards temporal information, and proposed **MoHI**, a moment-based spiking-to-analog handoff equivariant to affine time warps; perfect accuracy where rate readout stayed at chance (DVS-Gesture, N-MNIST, CIFAR10-DVS).
 - **Multi-View Fusion Pilot (manuscript in preparation).** Showed that naive cross-view max-fusion recovers recall but inflates false positives on normal studies; established the tradeoff as intrinsic at the score level.
 - Pre-register kill criteria and decision gates before committing GPU budget; IRB-approved clinical data collection with pediatric radiology collaborators at Riley Hospital for Children.

INDUSTRY EXPERIENCE

- **Augmedix**  July 2022 – July 2025
Software Engineer Level 2 — Backend and ML Infrastructure Dhaka, Bangladesh
 - Fine-tuned large language models for clinical note generation using **LoRA/PEFT**, and built the evaluation loop around them: factual-consistency and hallucination checks comparing generated documentation against source encounter transcripts.
 - Engineered scalable **Python and SQL pipelines on AWS** carrying high-throughput clinical audio and NLP streams into real-time generative documentation; served models through internal **FastAPI/Django** endpoints.
 - Re-architected and migrated legacy services onto **Kubernetes** with CI/CD, monitoring dashboards, and A/B rollout infrastructure, partnering with ML, backend, and clinical-operations teams on each deployment.
- **IDEEZA**  Sep 2023 – July 2025
AI Engineer — Generative AI and Computer Vision Remote

- Trained reward models and applied preference optimization to align generative outputs with human-rated quality, and designed **LLM-orchestrated workflows** routing a request across several generative models into one deliverable.
- Engineered multimodal image-to-3D reconstruction pipelines on **Stable Diffusion and ControlNet**, deployed at dev-ai-mesh.ideeza.com, with automated quality control scored by FID, SSIM, and LPIPS rather than manual review.
- **RadAssist**  May 2022 – July 2025
Research Engineer — AI Systems Dhaka, Bangladesh
 - Led R&D on a **multi-label CNN classifier** identifying 17 distinct radiological findings from chest X-rays, including per-class threshold selection and confidence calibration.
 - Instrumented evaluation pipelines with confusion-matrix analysis, calibration assessment, and systematic failure-mode review; deployed NLP summarization models translating structured predictions into readable clinical narratives.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=UNDER REVIEW, P=IN PREPARATION, T=THESIS

- [S.1] Alfaj Uddin Ahmed, et al. (2026). **A Benchmark Score Is Not a Property of a Model: Failure Composition Reorganizes Across Operating Regimes**. Under review at *ICLR 2027*.
- [P.1] Alfaj Uddin Ahmed, et al. **Why Models See Mirages: An Evidence-Starvation Theory of Ungrounded Multimodal Reasoning, and TIDE, a Training Objective That Provably Removes It**. Manuscript in final preparation for *CVPR 2027*; OpenReview preprint forthcoming.
- [S.2] Alfaj Uddin Ahmed, et al. (2026). **Temporal Moments at the SNN-ANN Interface: What Rate Readout Loses in Event-Based Vision**. Under review at *ICASSP 2027*.
- [C.1] Alfaj Uddin, et al. (2020). **IoT Based Cost Efficient Muscle Stimulator for Biomedical Application**. In *2020 IEEE Region 10 Symposium (TENSYP)*, 05–07 June 2020, Dhaka, Bangladesh. IEEE Xplore. DOI: [10.1109/TENSYP50017.2020.9231015](https://doi.org/10.1109/TENSYP50017.2020.9231015)

SELECTED RESEARCH AND OPEN-SOURCE PROJECTS

- **RAG-Based Clinical Note Summarizer** Open Source 
LangChain, LLMs, Vector Databases, Python
 - Engineered a **retrieval-augmented generation** pipeline that retrieves domain-specific clinical context at query time and conditions an LLM on it, reducing reliance on parametric recall for factual content.
- **Generative Fill Adapter — Semantic Inpainting with Diffusion Models** Open Source 
PyTorch, Stable Diffusion, ControlNet, LoRA, HuggingFace PEFT
 - Conditioned Stable Diffusion on semantic segmentation masks through a **ControlNet-style adapter** fine-tuned with LoRA, and compared conditioning strategies in an ablation scored on FID, SSIM, and LPIPS.
- **Self-Supervised Representation Learning for Sequential Signals** Open Source 
PyTorch, Transformers, Contrastive Learning, Masked Modeling
 - Pretrained **transformer encoders** on unlabeled 1D signal corpora with a combined contrastive and masked-reconstruction objective, then evaluated the learned representations on downstream sequence classification under transfer.
- **Cloud-Native ML Training and Serving Pipeline** Open Source 
AWS SageMaker, PyTorch DDP, Docker, FastAPI, REST
 - Architected multi-GPU training with **PyTorch DDP** and hyperparameter sweeps on SageMaker, then served the resulting classifiers from containerized FastAPI endpoints with logging and monitoring attached for reproducibility across environments.

LEADERSHIP AND SERVICE

- **European Rover Challenge — Sub-Team Lead**, *The European Space Foundation* 2019
Led a hardware sub-team through design and integration for an international Mars-rover competition; team placed 16th globally.

AWARDS AND ACHIEVEMENTS

- **Leslie A. Geddes Fellowship** — *Purdue University* June 2025
Competitive graduate fellowship awarded for academic and research excellence in biomedical engineering.
- **Kaggle Bronze Medal** — *RSNA Intracranial Aneurysm Detection* 2025
Placed 98th (top 5%) with a score of 0.69592 on a multi-class medical imaging classification task.
- **Design for Life — Ventilator Engineering Competition** — *2nd Place, Edge Foundation* July 2020
Built a functional mechanical ventilator from first principles under competition constraints.
- **IEEE SPS Winter School** — *Excellence Award, IEEE Signal Processing Society* Sep 2019
Recognized for excellence in an intensive multimodal signal processing program.

TECHNICAL SKILLS

- **LLMs & NLP:** Qwen3 and open instruction-tuned models, prompt design, reasoning-mode evaluation, LoRA / QLoRA / PEFT fine-tuning, vLLM serving, question answering, RAG, clinical text processing, spaCy, NLTK, LangChain
- **Deep Learning & Generative AI:** PyTorch, HuggingFace (Transformers, Diffusers, PEFT, Accelerate), TensorFlow, transformers and attention, vision-language models, diffusion (DDPM, latent, ControlNet), VQ-VAE, GANs, contrastive / SSL, CNNs, ViT
- **Experimentation & Evaluation:** Experimental design, pre-registered ablations, binomial / Clopper-Pearson / bootstrap intervals, failure-mode and error-taxonomy analysis, judge-based evaluation, parser auditing, calibration, mAP / AUC / sensitivity-specificity, FID / SSIM / LPIPS / KID, reproducible pipelines, automated test suites
- **Programming:** Python (primary), C/C++, SQL, Bash, MATLAB, JavaScript
- **ML Systems & Infrastructure:** Distributed training (DDP, DeepSpeed), **H100/A100 clusters**, CUDA, **AWS** (SageMaker, EC2, S3, Lambda), **Docker**, **Kubernetes**, FastAPI, Flask, Django, PostgreSQL, Firebase, Git, GitHub Actions, CI/CD, Linux